CS3010 – Computer Vision

DR. GAURAV SONI
SR.ASSISTANT PROFESSOR
SCHOOL OF COMPUTING SCIENCE ENGINEERING

Computer Vision

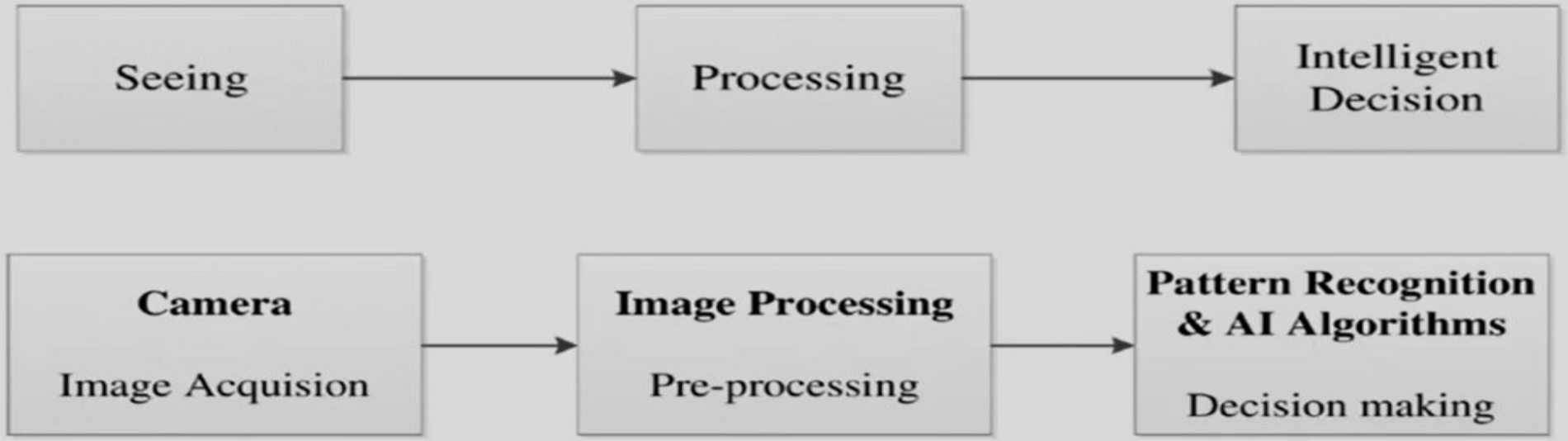
- Interdisciplinary field – computers gain high level understanding of digital images or videos
- Automate the tasks what human visual system do.
- Methods for image acquisition, processing, analysis are included
- Extract high dimensional data from real world

Definition of Computer Vision

- Building machines that see
  - Machines can interact with world.
  - Input is digital images and videos.
- Modeling biological perception

Computer Vision is a field of computer science that works on enabling computers to see, identify and process images in the same way that human vision does and then provide appropriate output.

Human Vision Vs Computer Vision



Major Milestones

- 1959 – First digital image scanner
- 1963 – Larry Roberts – described process of deriving 3D information from 2D photographs
- 1966 – Marvin minsky – connect camera to computer
- 1980 – Neocognitron was built
- 1991 – 1993 – Multiplex recording devices introduced with video surveillance for ATM machines
- 2001 – first face detection framework was built
- 2009 – testing robot cars
- 2010 – Google released goggles
- 2010 – Facebook - Tag photos

- 2011 – confirm the identity of Osama Bill Laden
- 2012 – AlexNet – University of Tornoto
- 2012 – Googles brain neural network
- 2015 – Tensor Flow was launched
- 2017 – Apple iphone X
- 2018 – Alibabas AI model
- 2019 – Facial recognition app - Police officers search images through mobile app
- 2020 – Intel Xe graphics card was launched

Computer Vision Vs Computer Graphics Vs Digital Image Processing

- Computer Vision
  - Input : Image Output : Interpretation
    - Image analysis, Image Interpretation, Scene understanding.
- Image Processing
  - Input : Image Output : Image
    - Image recovery, reconstruction, filtering, compression, visualization.
- Computer Graphics
  - Input: Scene representation Output: Image

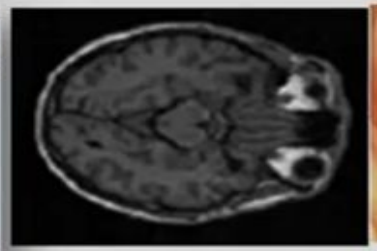
Applications of Computer Vision

Multimedia




Medical Imaging






Forensic and Biometrics



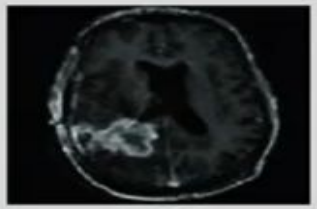

Video Surveillance

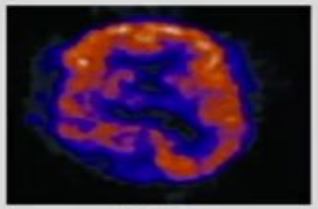


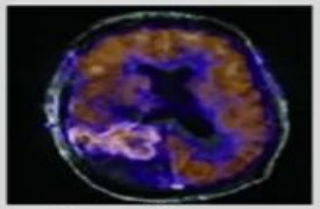
Remote sensing



Image Fusion


MRI


PET


Fused image


IR

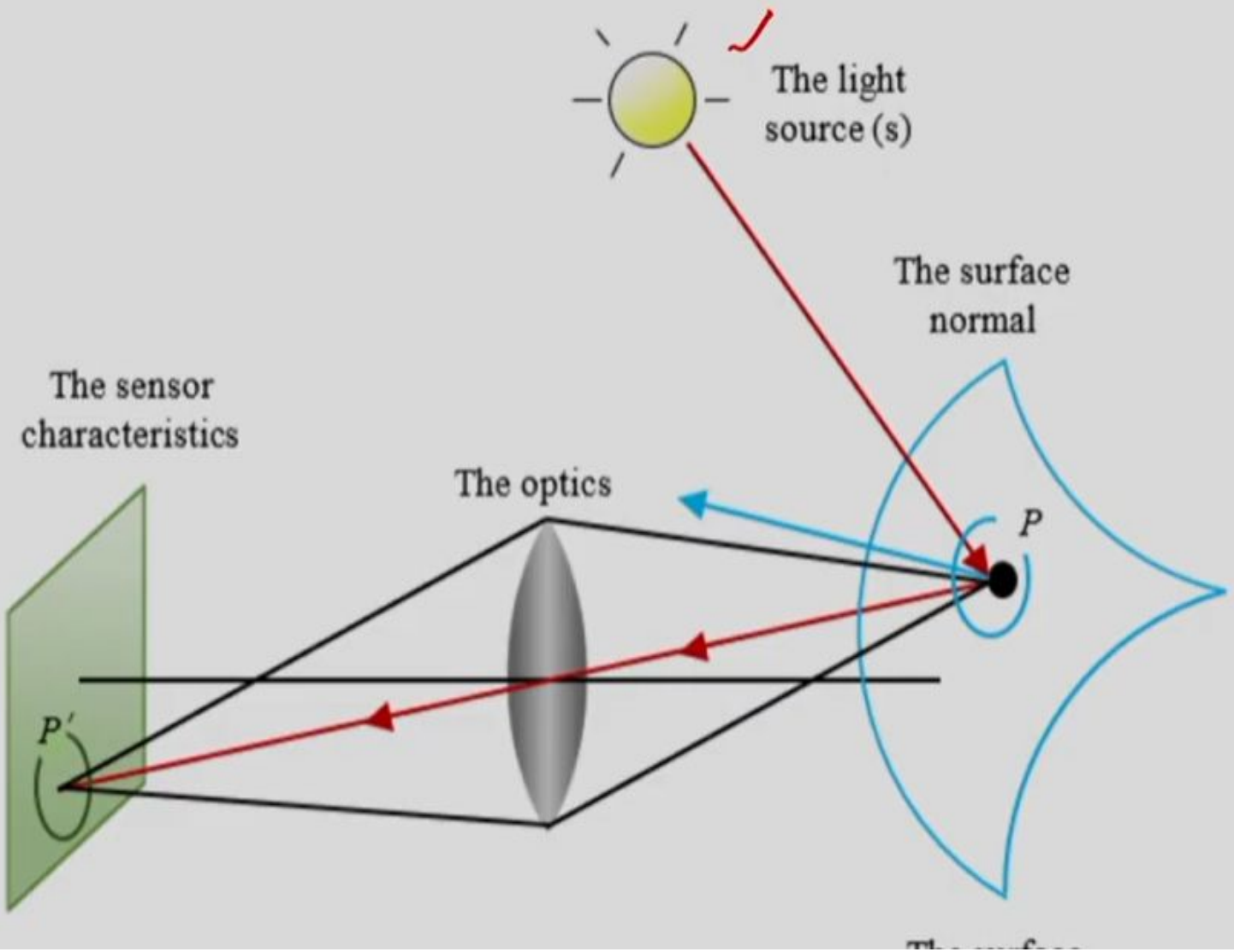

Visible


Fused image

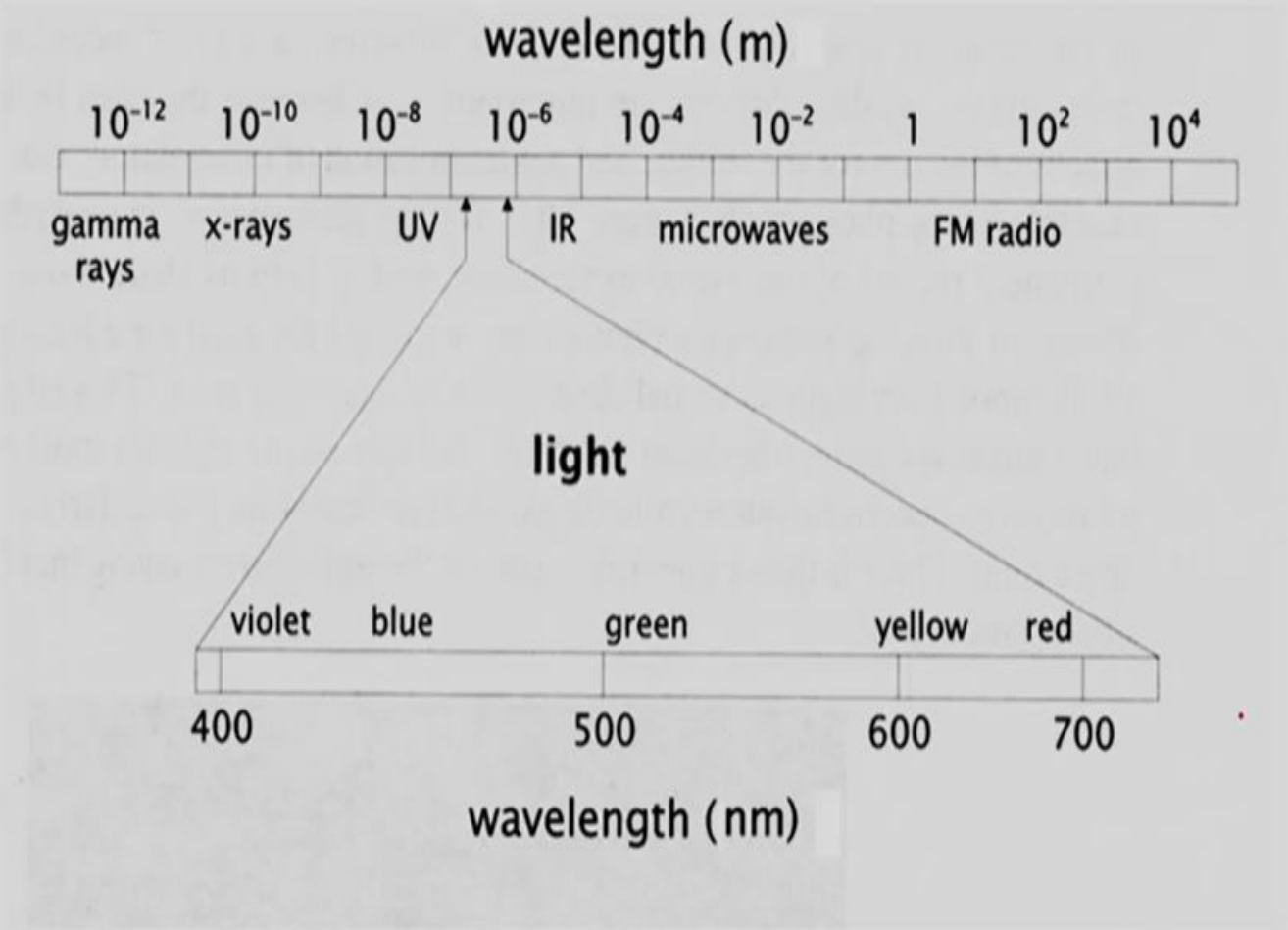
Applications of Computer Vision

- OCR
- Vision Biometrics
- Smart Cars
- Medical Imaging
- HCI
- Mobile Robot Control
- Image Collections

Image Formation



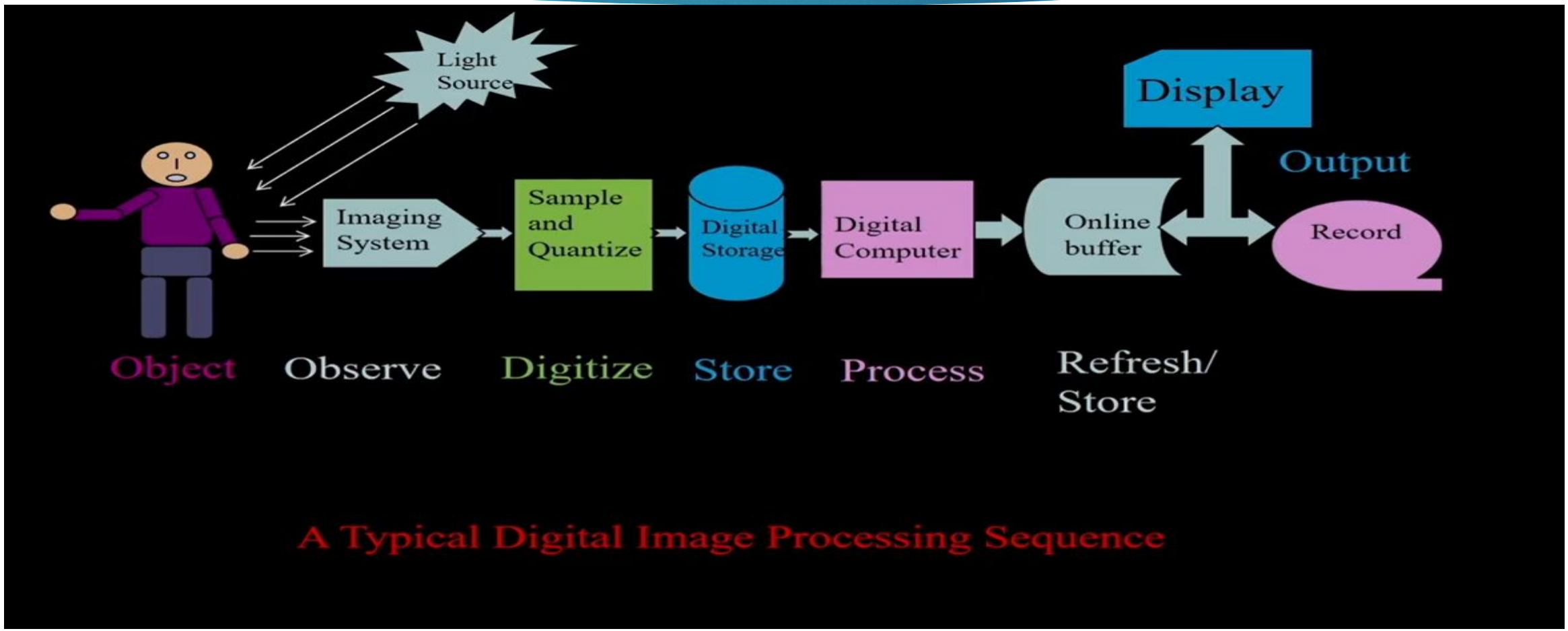
Source of Illumination



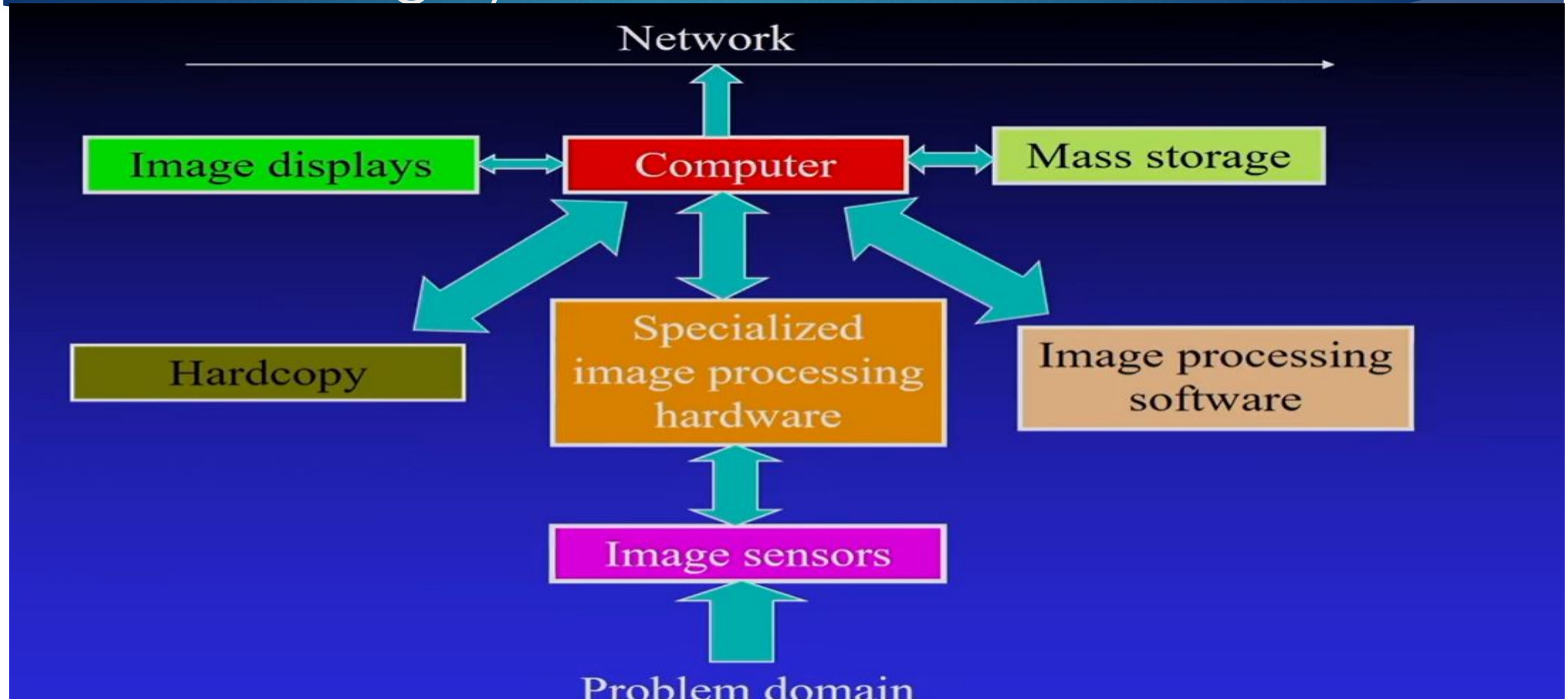
Types of Digital Images

- Digital photos.
- Image sequences used for video broadcasting and playback.
- Multi-sensor data like satellite images in the visible, infrared and microwave bands.
- Medical images like ultra-sound, Gamma-ray images, X-ray images and radio-band images like MRI etc.
- Astronomical images.
- Electron-microscope images used to study material structure.

Digital Image Processing



Components of General Image Processing System



Digital Image Processing

Processing of digital images by a digital computer

Need to process images?

Major applications that motivated

1) Improvement of pictorial information for human perception
2) Image processing for autonomous machine application
3) Efficient storage and transmission

Human Perception

Methods are employed that enhance pictorial information for human interpretation and analysis.

Applications include:

- Noise filtering
- Content enhancement
- Remote sensing
- Area of medicine
- Terrain mapping
- Atmospheric studies
- Astronomical studies

Noise filtering



Content enhancement



Content enhancement - deblurring



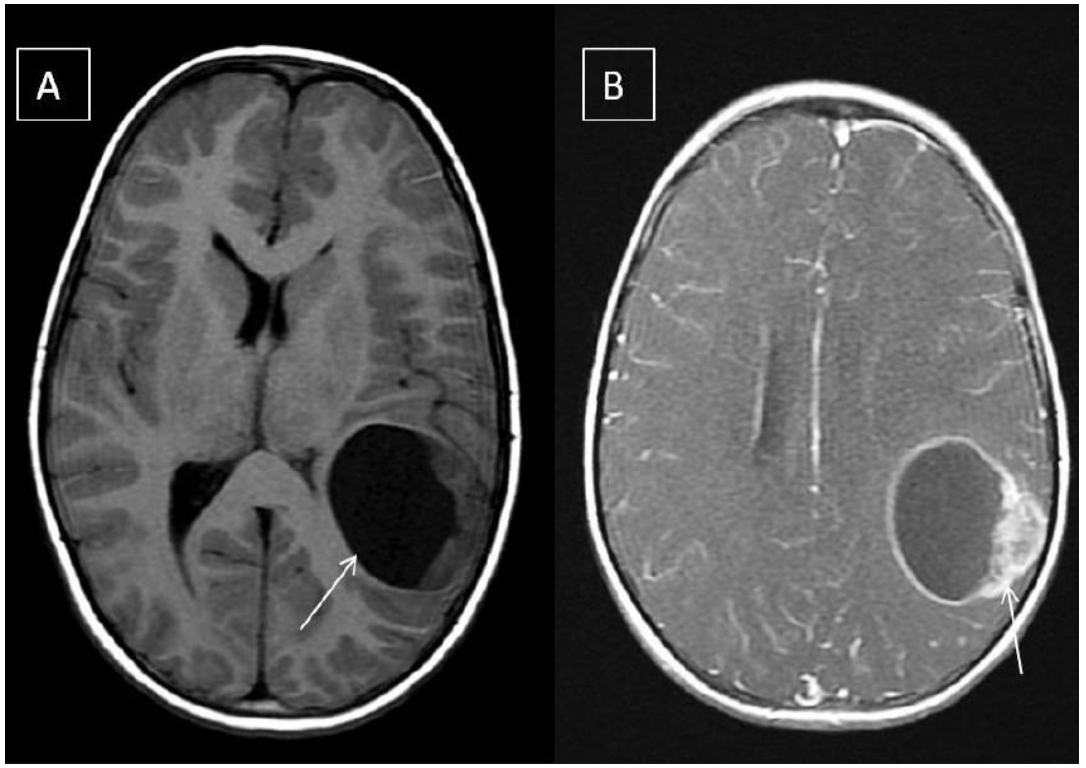


Remote sensing



Area of medicine





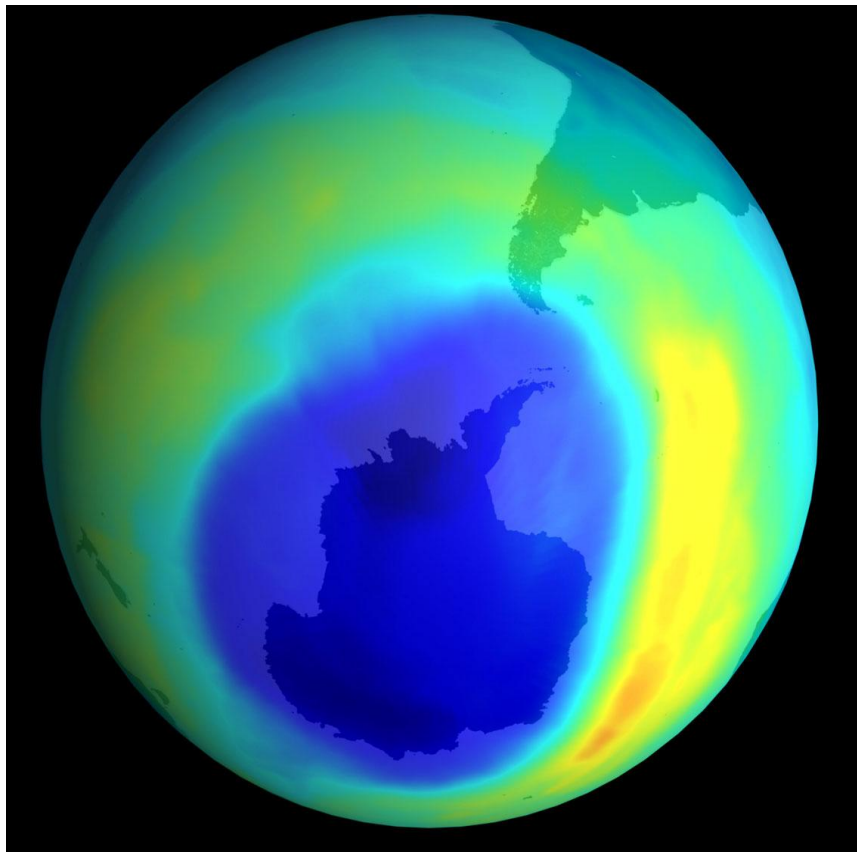
Terrain mapping



Weather forecast



Atmospheric studies - Ozone hole



Machine Applications

- Industry machine vision for product assembly and inspection
- Automated target detection and tracking
- Finger print recognition
- Machine processing of aerial and satellite imagery for weather prediction and

  crop assessment etc.

Image

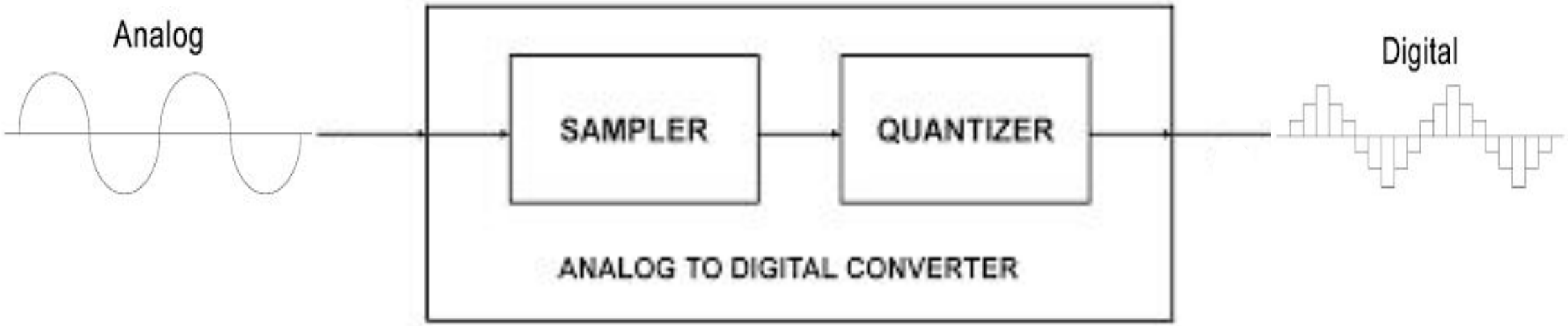
Image – Projection of 3D scene in 2D plane

A 2D function f(x , y), x , y □ spatial coordinates and f at any pair of coordinates (x , y) is called intensity or gray level of the image at that point.

Depending on x , y and f values, images can be classified in to

- Analog image
- Digital image

Analog – f (x , y) has continuous range of values.
Digital – f (x , y) has finite , discrete quantities.


Analog
SAMPLER
QUANTIZER
Digital
ANALOG TO DIGITAL CONVERTER

The Purpose of converting Analog to digital image with help of digital computers is to store and transmit efficiently.

Image contains finite number of elements.
Each element has a particular location and value.
These elements are called as picture elements or image elements or pels or pixels.

Advantages:

1) Fast processing
2) Cost effective
3) Effective storage
4) Effective transmission
5) Versatile image manipulation

Disadvantages:

High memory for good quality images and hence requires fast processing

Digital Image Processing

- Image processing is a method to convert an image into digital form
- Perform some operations on the converted image in order to get an enhanced image or to extract some useful information from it.
- It is a type of signal dispensation in which input is an image (video frame or photograph) and output may be image or characteristics associated with that image.
- Usually Image Processing system includes treating images as two dimensional signals and applying predefined methods on the images.

Low Level processes – involves primitive operations such as image pre-processing
Both input and output are images.
Mid Level processes – involves tasks such as segmentation , description and classification
Inputs are generally images but the outputs are attributes that extracted from the images
High Level processes – involves ensemble of recognized objects

Purpose of Image processing

The purpose of image processing is divided into 5 groups.

They are :

- **Visualization** - Observe the objects that are not visible.
- **Image sharpening and restoration** - To create a better image.
- **Image retrieval** - Seek for the image of interest.
- **Measurement of pattern** – Measures various objects in an image.
- **Image Recognition** – Distinguish the objects in an image.

Components of Digital Image Processing

Two elements are required to acquire digital images.

1. Sensor
2. Digitizer

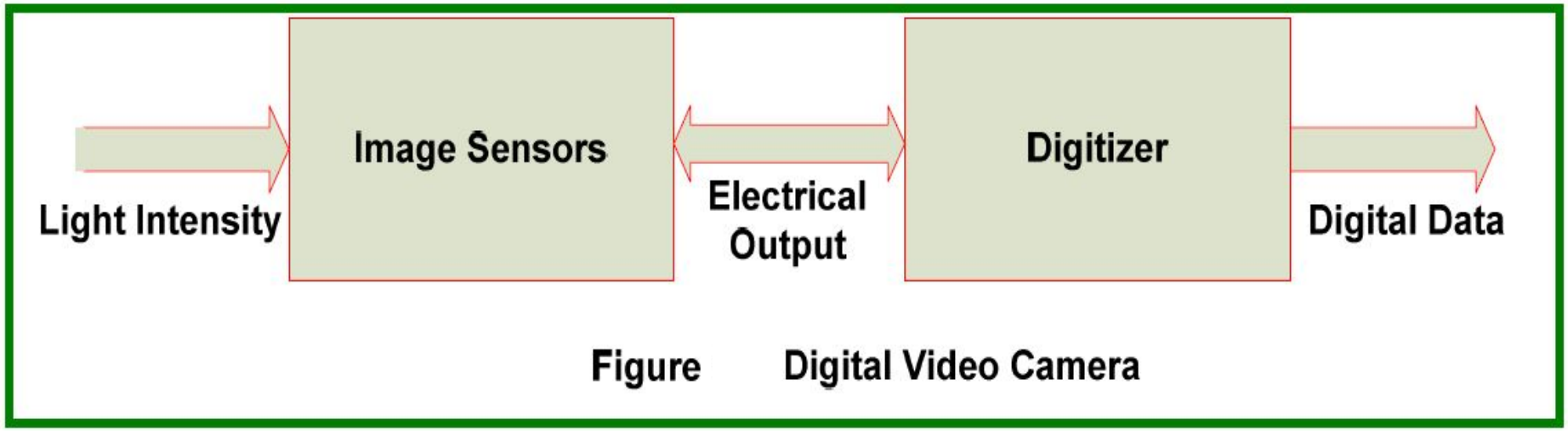


Figure Digital Video Camera